Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton Land Preparation

1.1 Soil Suitability, Site Selection, and Diagnostic Criteria

- **Soil suitability parameters:** Cotton (*Gossypium* spp.) can be successfully cultivated across a wide range of soil textures, including deep alluvial soils, sandy loams, loams, clay loams, and medium-heavy black soils. Optimal growth occurs in deep, well-drained soils possessing good moisture-retention capacity and aeration.
- **Unsuitable soil conditions:** Very sandy soils, highly saline/sodic soils, or waterlogged soils are unsuitable for cotton. Heavy soils prone to long-term water stagnation must be avoided unless artificial surface drainage is established.
- **Topography and slope:** Fields should be structurally uniform and flat to facilitate laser levelling, preventing localized water pooling or uneven irrigation coverage.
- **Soil testing and diagnostic sampling:** Soil samples should be collected from the 0–15 cm and 15–30 cm profile depths prior to field preparation and submitted to approved agricultural laboratories. Soil analysis determines pH, electrical conductivity (EC), organic carbon percentage, available macro-nutrients (N, P, K), and micronutrient statuses (Zinc, Boron, Magnesium).

1.2 Summer Deep Ploughing, Solarization, and Soil Health Recovery

- **Summer deep ploughing schedule:** Conduct deep ploughing during the hot summer months (May to June) immediately following the harvest of preceding Rabi crops. Deep ploughing is carried out using a tractor-operated Mouldboard (MB) plough to invert the topsoil layer.
- **Solar heat exposure and desiccation:** Leave the inverted soil exposed to direct sunlight for at least 2 to 3 weeks during peak summer heat.
- **Destruction of soil-borne pests and pathogens:** Summer solarization bakes the exposed soil layer, desiccating and destroying:
  - Overwintering pupae and larvae of insect pests, particularly bollworms (*Helicoverpa armigera*, *Earias* spp., and *Pectinophora gossypiella*).
  - Resting structures, sclerotia, and chlamydospores of soil-borne fungal pathogens responsible for fungal root rot (*Rhizoctonia* spp.) and Fusarium wilt (*Fusarium oxysporum* f. sp. *vasinfectum*).
  - Plant-parasitic nematodes in the upper root zone.

- **Perennial weed seed bank suppression:** Deep inversion buries weed seeds and exposes underground rhizomes, tubers, and bulbs of stubborn perennial weeds (such as *Cyperus rotundus* and *Cynodon dactylon*) to solar heat for complete desiccation.

1.3 Subsoiling, Hardpan Breaking, and Root-Zone Management

- **Subsurface hardpan formation:** Continuous cropping, shallow tillage, or intensive puddled paddy-wheat crop rotations create a dense, impermeable hardpan at 20 to 35 cm soil depth.
- **Physiological damage from hardpans:** Subsurface hardpans obstruct vertical taproot extension, causing roots to bend horizontally at right angles (right-angle root bending). Restricted taproots reduce nutrient and deep-soil water extraction, predisposing cotton plants to crop lodging, severe drought stress, and parawilt (sudden wilt complex).
- **Subsoiling operation protocol:**
  - Operate a tractor-drawn sub-soiler or chisel plough crosswise across the field at 1.0 m spacing prior to primary seedbed preparation.
  - Alternatively, execute deep ploughing using a Mouldboard (MB) plough to shatter subsoil compaction.
- **Agronomic benefits of subsoiling:**
  - Shatters dense subsurface hardpans without bringing infertile subsoil to the surface.
  - Enhances vertical water infiltration and subsoil moisture storage capacity.
  - Encourages deep, uninhibited vertical taproot penetration into subsoil layers.
  - Suppresses parawilt incidence and prevents wind-induced crop lodging during heavy boll loading.

1.4 Secondary Tillage, Soil Pulverization, and Clod Crushing

- **Clod breaking with Disc Harrow:** Following MB ploughing or subsoiling, operate a tractor-drawn Disc Harrow to cut through large soil clods and crop residues.
- **Soil pulverization with Cultivator:** Perform 2 to 3 rounds of cross-cultivation using a spring-loaded tiller or cultivator to pulverize the soil, breaking smaller clods and refining soil tilth into a friable seedbed.
- **Clod crushing and surface sealing with Planker (*Suhaga*):** Pass a heavy wooden or iron Planker (*Suhaga*) immediately after each cultivator operation. Planking crushes remaining surface clods, levels micro-relief, compacts loose soil slightly, and seals the upper soil surface to prevent evaporation loss.
- **Ideal seedbed characteristics:** The finalized seedbed must possess a fine, clod-free, firm, and well-aerated physical tilth that allows precise seed placement at 3 to 5 cm depth.

1.5 Pre-Sowing Irrigation (*Rauni*) and Moisture Conservation

- **Purpose of pre-sowing irrigation (*Rauni*):** In dry regions or irrigated canal zones, apply a heavy pre-sowing irrigation (*Rauni*) with good-quality canal water or non-saline tube-well water. This fully saturates the soil profile, providing moisture for uniform seed germination and early seedling establishment.
- **The "*Wattar*" workable moisture condition:**
  - Following *Rauni*, wait for the soil to dry naturally until it reaches the *Wattar* (workable moisture) condition.
  - *Wattar* represents the precise moisture state where soil is dry enough to support tractor machinery without causing compaction or structural destruction, yet moist enough to ensure rapid seed germination.
- **Tillage under *Wattar* condition:** As soon as the field attains *Wattar*, execute cross-cultivation with tillers 2 times, followed immediately by heavy planking (*Suhaga*). This seals surface capillary pores, conserving moisture inside the seedbed for sowing.

1.6 Precision Laser Land Levelling and Drainage Infrastructure

- **Laser land levelling protocol:** Level the dry field using a tractor-operated Laser Land Leveller prior to *Rauni* or basal bed formation.
- **Benefits of laser levelling:**
  - Achieves precise land topography, eliminating micro-high and micro-low spots.
  - Ensures uniform irrigation distribution across the entire plot, saving 20% to 30% irrigation water.
  - Prevents localized water pooling and uneven fertilizer leaching.
  - Facilitates uniform crop emergence and uniform plant stand.
- **Surface drainage layout:** Construct deep surface drainage ditches along field boundaries and lowest contours. Cotton is sensitive to water stagnation (>24 hours causes root oxygen starvation and wilting), making rapid evacuation of excess monsoon rainwater essential.

1.7 Basal Organic Soil Enrichment and Biological Amendments

- **Farmyard Manure (FYM) application:** Incorporate 5 to 10 t/ha (or 5 t/acre) of well-decomposed Farmyard Manure (FYM) or compost into the soil during initial secondary tillage (2–3 weeks before sowing).
- **Neem Cake incorporation:** Apply Neem Cake @ 250 kg/ha (or 5 q/ha) during field preparation in fields with a history of nematodes or subterranean termites. Neem cake improves organic carbon, inhibits nitrification, and suppresses soil pests.
- **Biological disease management (Biocontrol):**

  - Mix *Trichoderma viride* or *Trichoderma harzianum* (1% WP @ 10 kg/ha) with 200 kg well-decomposed FYM.
  - Moisten the mixture slightly and incubate under shade for 7 days to allow fungal spore multiplication.
  - Broadcast this bio-enriched mixture uniformly across the field during final land preparation before seedbed formation to suppress soil-borne fungal pathogens (*Rhizoctonia*, *Fusarium*, *Sclerotium*).

1.8 Field Layout, Seedbed Configurations, and Sowing Preparation

- **Flat bed layout:** Conventional flat seedbed prepared by levelling and planking. Suitable for light-to-medium loamy soils with canal or sprinkler irrigation.
- **Ridge and furrow layout:**
  - Form ridges and furrows using a tractor-operated ridge-furrow maker after final seedbed refinement.
  - Ridges facilitate furrow irrigation, improve root zone aeration, and prevent water stagnation around plant bases.
- **Narrow raised bed-furrow layout:** Form narrow raised beds using a bed former. Seeds are planted either on bed tops or inside the furrows based on soil moisture and salinity conditions.
- **Broad raised bed-furrow layout:**
  - Construct broad raised beds using broad bed formers.
  - Sowing is conducted on one side or both sides of the broad bed top. Broad beds save water, improve drainage during heavy rains, and reduce soil crusting.
- **Special precautions for saline soils:** In salt-affected soils or when using moderately saline irrigation water, sow seeds on flat beds or inside the furrows of ridge systems rather than on ridge tops, as capillary evaporation concentrates harmful salts on ridge crests.

1.9 Integrated Step-by-Step Land Preparation Workflow

1. **Pre-Season Summer Phase (May–June):** Execute deep summer ploughing using an MB plough. Leave soil exposed to sun for 2 to 3 weeks for solar desiccation of pests, pathogens, and weed tubers.
2. **Hardpan Breaking Phase:** Cross-subsoil with a chisel plough at 1.0 m spacing to break subsurface hardpans and improve deep taproot penetration.
3. **Levelling and Soil Enrichment Phase:** Level the field using a Laser Land Leveller. Broadcast 5–10 t/ha FYM, 250 kg/ha Neem Cake, and *Trichoderma*-enriched FYM, then incorporate using a disc harrow.

4. **Pre-Sowing Irrigation Phase (*Rauni*):** Apply a heavy pre-sowing irrigation with canal water to charge the soil profile.

5. **Final Seedbed Refinement Phase:** Wait for the soil to reach *Wattar* moisture condition. Cultivate 2 times with a tiller followed immediately by planking (*Suhaga*) to seal moisture.

6. **Field Layout Phase:** Prepare flat beds, ridges-and-furrows, or raised beds using tractor-operated bed/ridge makers based on zonal recommendations and irrigation infrastructure.